fMRI meta-analysis

—Activation Likelihood (ALE)

pre：MYL

步骤
STEPS

1. 提取foci
2. GingerALE
3. 坐标转换
   (MNI↔Talairach)
4. 单数据分析
5. 其他分析
6. mango

提取foci

纳入标准：

- 使用标准坐标空间(MNI/Talairach)
- 排除ROI分析，选择whole-brain analysis
- ANOVA只保留main effect(最好标注了方向)
- 注意：ALE分析最小数据量为15
  (最好17以上)

Contrast	Brain area	x	y	z	T-value	k
Arithmetic–letter reading	Left PSPL	−28	−72	42	5.80	664
	Left hIPS	−32	−58	42	5.58	
	Right PSPL	32	−70	42	5.13	377
	Right IFG	42	20	−2	4.80	282
	Right hIPS	48	−40	48	4.06	200
Multiplication–subtraction	Left STG	−60	−14	0	5.87	1616
	Right STG	60	−2	−2	5.53	1047
Finger task–colour task	Left PSPL	−20	−68	62	11.61	5914
	Left hIPS	−38	−44	50	11.08	
	Left EBA	−48	−68	−2	7.03	
	Right hIPS	44	−40	52	10.87	5966
	Right PSPL	30	−72	40	9.16	
	Right EBA	54	−60	−8	7.06	335
	Left FEF	−28	−4	50	6.37	681
	Right IFG	38	26	−4	6.35	534
	Right FEF	26	−2	56	5.08	874
	Right MFG	50	32	30	4.65	548
	Left PrCS	−46	4	34	3.97	283
Right hand–left hand	Left M1/ S1	−46	−22	62	7.39	895
Left hand–right hand	Right M1/ S1	42	−20	62	6.46	188

提取foci

提取形式及注意：

- 第一行列出第一作者的姓名、年份，实验名称或被试组名称。
- 下一行是这一组焦点的受试者数目
  随后是坐标数据。
- 所有非坐标数据应该放在评论中，也就是"//"之后。
- 一定要在不同组的焦点之间留有一个空行，
  但同一组的焦点之间不要有空行
- x、y、z坐标之间相隔一个制表符

```
// Reference=Talairach
// Hui, 2000: Acupuncture vs. Tactile Stimulation, Increases
// Subjects=13
56          -15         50
-59         -18         43
59          -21         21
-50         -18         18

// Li, 2003: Conventional Acupuncture > Rest, Activations
// Subjects=20
-5.12       -82.33      10.8
2.3         -77.71      11.37
15.36       -62.35      7.64
45.18       15.75       15.54

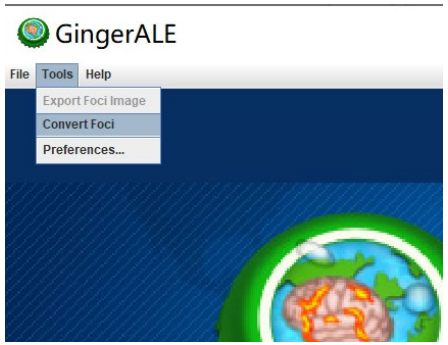
// Li, 2003: Electro-Acupuncture 2 Hz > Rest
// Subjects=20
2.28        -77.88      13.15
-6          -68.35      12.11
-56.89      -63.43      11.72
49.74       15.21       20.97
```

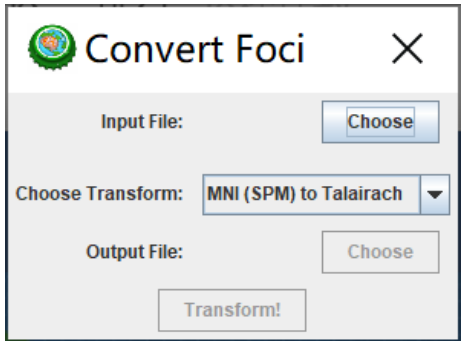
坐标转换

将所有坐标统一，(MNI↔Talairach)

使用GingerALE进行坐标转换(brainmap 官网下载)

- **注意更改output File**






```
// Reference=MNI
// Chochon ,1999: Multiplication  vs.  Digit  Naming
// Subjects=8
56.67       8.89        37.27
33.61       81.58       23.75
50.01       43.66       30.58
53.44       38.53       44.48

// Reference=MNI
// Chochon ,1999: Multiplication  vs.  Comparison
// Subjects=8
33.71       79.01       30.72
```

GingerALE

BrainMap software论坛答疑:

http://www.brainmap.org/forum/viewforum.php?f=3&sid=bb8f99114a084404469109b7d5383f12

1. Q：contrast出现A>B, B>A的情况该如何提取？

A：如果只分析A>B，则只提取A>B，Combining them (or choosing only one to include) is best practice. Leaving them separate causes GingerALE to treat them as independent data points, but they are the same subjects and aren't really independent.

2. Yes. If you had studies that were solely studying patients and others solely studying controls, you could use GingerALE's contrast analysis and it would give you P>C and C>P. But since you have studies that are showing results of Patients > Controls already, it's best to group those together to try to find agreement between them, then separately compare Controls > Patients.

2. Q：如果实验组与控制组人数不相等，experimental>control的情况subject该输入多少？

A：为了避免假阳性，选择人数更少的那组填写。

When entering the number of participants in a study, what does one do if there were different participants in different contrasts? For instance, if there were three groups being contrasted, with different N's in each group. Should one make a unique input for each of the contrasts within the study, something like this?
For illustration, lets assume that A =30, B=25, C =20

We recommend using the size from the smaller of the groups being compared. The subject size is used to compute the amount of blurring and uncertainty around the coordinates. With smaller subject groups, there is more uncertainty. So when comparing two groups, we recommend being conservative and using the uncertainty from the smaller of the groups so you don't over estimate the strength of the results.

So in this case:

// exp 1 A>B
// Subjects=25
coordinates

// exp 2 B>C
// Subjects=20
coordinates

// exp 3 A>C
// Subjects=20
coordinates

GingerALE

BrainMap software论坛答疑:

3. ANOVA只保留main effect (标注了方向)

Unfortunately, there isn't anything like an interaction-analysis in GingerALE at the moment. Out of my head, the only way I could imaging to implement that would be to compute the interaction-term on the ALE-maps [1 -1 -1 1] and then use a permutation procedure similar to what is performed now in the contrast - however, this time the pool of experiments would need to be divided (according to the original proportions) across the four sets. Conceptually this isn't a fundamental problem. However, I would expect, that with a rather limited number of studies, statistics would be on rather thin ice.

Another approach you could take is to perform a ALE-contrast between the ("old vs young" contrasts on task 1) and the ("old vs young" contrasts on task 2). This would effectively test for differences in differences.

4. Q.:一个研究中的不同对比(A>B、A>C)，都要纳入，如何排列？

A：取决于是否视为独立组，如果独立//中间要空行，不独立则为右图

- Different studies, and using different sample, then arrange as in the 1st example.
- Multiple experiments in a study, therefore using same sample, then arrange as in the 2nd example.
- Different studies (e.g. AuthorXX, 2004; AuthorYY, 2003), but using same sample, then arrange as in 2nd example.

2nd example:

```
// Reference=MNI
// Chochon ,1999: Multiplication vs. Digit Naming
// Subjects=8
56.67    8.89    37.27
33.61    81.58    23.75
50.01    43.66    30.58
53.44    38.53    44.48
// Reference=MNI
// Chochon ,1999: Multiplication vs. Comparison
// Subjects=8
33.71    79.01    30.72
```

liikii wrote:

Hi, Following up on the question posted by Briandh, do we need to do anything to the text file to control for multiple experiments from a single study, after separating the contrasts as if they are from independent studies, assuming I want to put them in a single meta-analysis?

Hi liikii,
The first example you give, GingerALE will treat as independent foci groups. In the second example, GingerALE will treat as a single group.

If you have already separated out independent contrasts or experiment groups, then you don't need to edit the text file. So if they are independent, they should be separated as in the first example.

If there are non-independent contrasts, but you want to include them all anyway, you can join them together as in the second example. If you are using Sleuth, you can also use the "Export as Subject Groups" option, which will export contrasts joined together by subject group, sort of like your second example.

-Mick

In contrast, in another recently published paper (https://www.biorxiv.org/content/early/2018/02/15/265496), the authors wrote "Studies entered into the CBMA must be independent to avoid bias. Coordinates from multiple experiments in single studies using the same subjects were therefore pooled into single independent studies." and "Different studies sharing subjects were pooled to form a single independent study." in the "analysis" section. According to my understanding, this means that all coordinates survived statistical thresholds in the same sample were pooled into a single independent study (experiment) , which is inconsistent with the method used in the first study.

GingerALE

BrainMap software论坛答疑:

5. 关于threshold;

Q:How to best choose threshold?

A:FWE如下，两个阈值，p和volume-size。

建议使用未校正的p进行cFWE

Which threshold method did you use? I think it's the line immediately previous to "Threshold Value = 0.05".

- If it was Cluster-Level FWE, then it is a two-fold threshold. First a cluster-forming threshold is applied, usually an Uncorrected P value threshold, to find clusters (contiguous regions). Then a volume-size threshold is applied to remove the smallest clusters. It chose the largest 5% (from Threshold Value 0.05) of 1000 permutations of randomized data to find the 50th largest results that occurred by chance and exclude all volume sizes that were smaller than that.
- If it was Voxel-Level FWE, then that is a simpler threshold which in practice is very strict. It finds the distribution of maximum ALE values across randomized data and thresholds the ALE images directly. It would find the 50th largest ALE value (from your Threshold Value & Permutations settings) from chance and exclude all values below that.

My suggestion for choosing a threshold in GingerALE is to first run a few simple P value thresholds. The simple P value thresholds are very quick to run since they are not using any permutations.

P < 0.001 is often recommended. However, ALE values and P values depend heavily on the dataset size - P < 0.001 won't be appropriate for all studies. If you are investigating something without many publications available, you may have a smaller dataset and won't get anything informative out of such a strict threshold. Also, if you are doing a large-scale analysis with several hundred experiments, P < 0.001 will not be conservative enough and will result in very large regions that cover too many distinct regions of the brain which will be difficult to interpret.

So, I would also try P < 0.01 and 0.05 for small datasets. Or < .0001 (or smaller!) for very large datasets. The goal is to choose a P value that gives reasonably-sized clusters, roughly more than 50mm^3 and smaller than 5000mm^3. (For really, really large datasets it is hard to find a small enough P value that works and you should use voxel-level FWE instead, but this is a nice problem to have.)

Then, use that P value as your "cluster-forming threshold" within the cluster-level FWE. Cluster-level FWE is the most generally recommended threshold. It uses randomly generated datasets and many permutations to find out what size of cluster can result from chance / noise when using that P value. It then sets the minimum cluster size to remove small clusters that are likely noise. Be careful with using too few permutations, as that can sometimes cause/facilitate the chosen minimum volume to change when re-analyzing the same data.

So, I'd recommend 0.01 (or maybe 0.05) clustFWE with at least 1000 permutations and a cluster-forming threshold of the P value threshold you found above that fits your dataset.

@dannick Yes, in general I'd recommend cFWE < .05 using a cluster-forming threshold of p < .001 or .0001 or even smaller. The cluster-forming threshold gives you some flexibility in adjusting your threshold.

Remember that the cluster-forming threshold is applied first, then the permutations are used to choose a minimum volume threshold which would remove entire clusters for being below the size that could occur by chance. So in order to see what cluster volumes we're working with, you can run an analysis using Uncorrected P value as the threshold. If the result is a single very large cluster that runs across several distinct regions of the brain, try making your cluster-forming threshold more strict in order to break that up into separate regions. You could open up the P value image in an image viewer to get a feel for a good threshold, or you could run a few P value settings (.001, .0001, .00001).

ALE values and P values will depend on the statistical strength of your dataset. Larger datasets can get much more statistical significance. Adjusting the cluster-level threshold like this gives you a chance to compensate for the particular size of your dataset and to get cluster sizes that are easily interpretable.

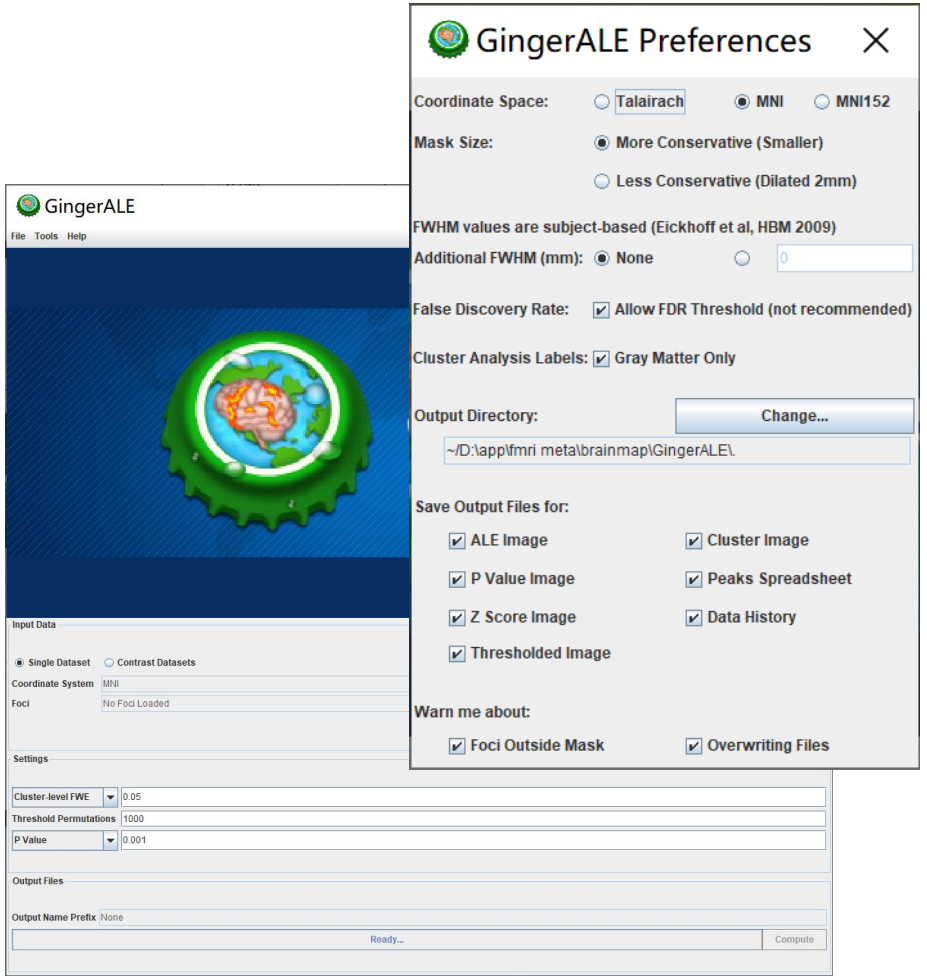
As the dataset gets larger and the P values get smaller, eventually it'll be hard to choose an appropriate cluster-forming threshold. This is when I think vFWE really shines. It is a particularly conservative threshold and so it fits well with a large dataset. vFWE does not use a cluster-forming threshold, but instead uses the permutations to get the single maximum ALE value from each randomized dataset. So vFWE 0.05 using 1000 permutations would choose the 50th largest ALE value out of the permutations. This ALE value is used as a threshold on the original ALE image. You can use a (user defined & fairly arbitrary) minimum cluster volume as well when you are using vFWE, but you don't have to. I'd recommend setting it to something small, like 0 or 50 or 100, since the vFWE threshold is already very strict you don't need to further reduce it much.

@alexpuiu Chumbley & Friston, NeuroImage 2009 suggest that voxel-wise False Discovery Rate thresholds are not appropriate for use on smooth data that was created with Gaussian curves. Since ALE analyses are using Gaussian smoothing and in cFWE the cluster-forming threshold is applied voxel-wise to create the cluster regions, then FDR is not a great fit for this.

单数据分析

步骤：

- GingerALE 菜单中File→Open
- 这里选择cFWE分析
- 输出：tool →preference可以选择输出位置；
  输出文件格式在GingerALE手册中
  - ALE图像: *_ALE.nii
  - P值图像: *_P.nii
  - 矫正后的ALE图像: *_ALE_pID05.nii
  - cluster图像: *_clust.nii
- 要进行下一步两个meta的对比分析或联合分析则要先分别进行两次单数据分析。结果可在输出表格中查看激活脑区，也可选择校正后的ALE图像(红框)，在mango中进行可视化解读(ppt最后一部分)



其他分析——联合分析和对比分析

步骤：

- 分别进行两次单数据分析后
- 合并两个任务坐标点(可以直接复制)，使用GingerALE 菜单中File-Save &Merge Foci 选项来合并焦点
- 依次放入三个dataset
- 联合分析：”A_conj_B_ALE.xls”
- 对比分析：

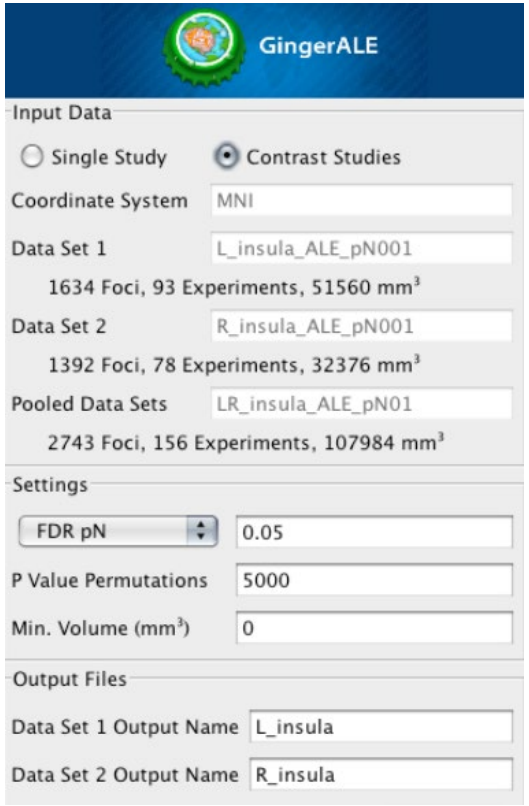
未矫正的ALE图像: *_ALE.nii

P值图像: *_P.nii；

矫正后的图像(由ALE值转换成的z值图像):如*_Z_pID05.nii；

cluster图像：*_cluster.nii

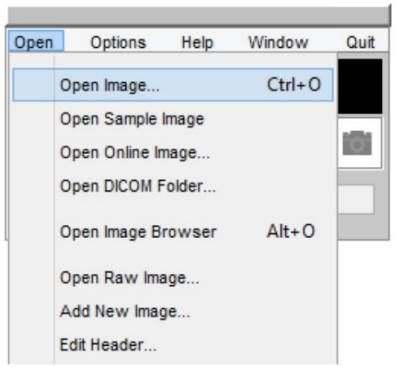
文本文件(.xls, .txt)。



可视化

步骤：

- 使用mango进行可视化
- 在brainmap官网可下载MNI和Talairach模板图(http://brainmap.org/ale/)
- 先打开模板图像，再打开GingerALE校正后图像结果



- file →Add Overlay，选择校正后的ALE图像
- Edit →update to Image Range



可视化

步骤：

- 可利用matlab内置插件xjviewer辅助解读结果

